

Night Buddy

You Sleep. Someone's Got You.

Meta Sponsor Challenge Specification

Companion to the Irin Hackathon Project Specification and the Irin Rounds Specification

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Night Buddy is Irin's entry concept for the Meta challenge: a peer safety network where adults with type 1 diabetes watch over each other's nights. It is designed to ride on machinery already being built for core Irin and Irin Rounds (alarms, presence, the relay, pairing, the narrative pipeline), so most of this document specifies design rather than new construction. Build sequencing is strict: Night Buddy code starts only after core Irin works end to end and R1-R11 landed with a full day left; otherwise Night Buddy appears in the Meta submission as a staged concept in the demo video. All CLAUDE.md repo rules apply.

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1. The Sponsor Challenge and Its Deliverables

Meta challenge, "Bringing People Closer Together with AI." Mission, verbatim:

- Build an AI-powered social product that helps strengthen connections in new ways.
- Use AI in a meaningful way.
- Create a working prototype and a 2 to 3 minute demo video showing the project in action.
- Submit a public code repository and a short write-up explaining who the project is for, how it strengthens connection, and why AI is essential.

Judged on: how meaningfully the project strengthens human connection; how essential and well integrated AI is; originality; and the strength of the working demo. Prize: top 3 teams advance to a Round 2 hackathon at Meta's Menlo Park campus, travel and accommodations provided.

Note the deliverables carefully, because two of them are work items, not code: the 2-to-3-minute VIDEO and the WRITE-UP. The public repository is a fresh mirror built on submission day by `deploy/public_mirror.sh` (squashed history, journal/ excluded); the working repo stays private. The video is assigned to Slavik, who has the most slack after hardware assembly.

2. Terminology

Term	Meaning
Buddy / pairing	Two consenting T1D adults linked as mutual overnight backstops. The link uses the same key-exchange pairing machinery as Irin Rounds (QR, keys exchanged, PIN-confirmed, revocable).
Twin buddy	A buddy in roughly your own time zone. Shared daily rhythm, strongest human-connection mode, and (with two devices) mutual room-scale waking.
Mirror buddy	A buddy roughly 10 to 14 hours away. Your night is their afternoon, so your watcher is awake with their phone in hand. Safety-optimal: it removes the "will the backstop wake up" problem entirely, and it is still perfectly reciprocal, since each covers the other's night during their own day.
Watcher / watchable	The two sides of the network. A watcher responds to alerts; a watchable person can generate them. Everyone on the network is watchable; watching is opt-in.
The ladder / a rung	The escalation ladder is the ordered chain of increasingly loud responses to a low. Core Irin already has rungs (warning, full alarm, escalation, re-arm). Night Buddy adds one final rung: another human being.
The hub	An opt-in, community-wide list of currently unacknowledged severe lows that verified T1D volunteers anywhere in the world can see and act on. Section 7.
Listing	One entry on the hub: a pseudonymous, content-light record of an active unacknowledged low ("Sam G., alarm unacknowledged 14 min, presence confirmed").
Claim / lease	A volunteer taps "I've got this" and gets exclusive responsibility for a listing for a fixed window (the lease, starting value 3 minutes). Prevents ten people

	calling at once. If the lease expires unresolved, the listing reopens.
Treating status	The Brain-only acknowledgment: one giant button ("I'm treating") on the phone alert that clears the person from the hub and tells any claim-holder they are awake and handling it. It expires (starting value 20 minutes); if glucose has not recovered, the listing returns at higher urgency.
Resolution by glucose	A listing closes when the CGM shows recovery (or the treating window resolves clean), never merely because a volunteer says they called. Data closes the loop.
Brokered contact	Volunteers and patients never see each other's phone numbers. The relay places or connects the call (or triggers the patient's device chime). No personal contact details cross the network.
Emergency script	The patient's own pre-written instructions ("call my mom at ...", "my neighbor Ted has a key"), shown only to the current claim-holder, only while the claim is live.
PulsePoint / GoodSAM	Existing, proven volunteer-responder apps that alert CPR-trained citizens to nearby cardiac arrests. Night Buddy's hub is that pattern applied to unacknowledged lows, with "nearby" replaced by "awake," since the intervention is a phone call.
Web Push	The browser standard for sending notifications to a phone from a web app. Not part of the build (decision 29): it cannot ring through a locked, silenced phone, so the demo's second channel is the WhatsApp message and production is a native app with Critical Alerts or a call; mirror matching exists for the same reason.
Brain-only	A participant with a CGM and the phone app but no bedside device. Full network member; weaker wake chain and lower-confidence listings, labeled as such.
Dead in bed	The community's name for the fear this product addresses: a severe overnight low, alone, with nobody to hear an alarm.
(From the other specs)	CGM, T1D, low, level 2 low (<54 mg/dL), near-miss, ack, re-arm, presence, relay, PIN gate, clock.py, DEMO badge: all defined in the Irin and Rounds specifications and used identically here.

3. What Night Buddy Is

3.1 One sentence

Night Buddy pairs adults with type 1 diabetes as mutual overnight backstops and, at scale, gives the whole community an opt-in watch list, so that when someone's low alarm goes unanswered at 2 AM, another person who knows exactly what a low feels like makes sure they wake up.

3.2 The problem

A severe overnight low can leave a person unable to wake themselves or call for help. Every existing safety net assumes someone else is in the house: a partner who hears the alarm, a parent down the hall. For T1D adults who live alone there is no someone, and the existing sharing tools make it worse, not better: Dexcom Follow means asking a friend to watch your raw medical data stream around the clock, forever, which is such a heavy ask that many people living alone simply have nobody watching. The fear does not go away. The community has a name for it: dead in bed.

Now look at the shape of the problem: millions of strangers who share the same condition, the same fear, and complementary schedules, each of whom needs exactly what every other one can offer. That is a matching problem. Which makes the solution a social product, and squarely Meta's brief: AI that sparks new connections from common ground. The common ground here is not a hobby. It is mutual survival.

3.3 What it does, in five sentences

1. You opt in and are matched with another T1D adult, a twin (your time zone) or a mirror (the other side of the world, awake during your night).
2. Ninety-nine percent of nights, the entire relationship is one AI-written morning message to each of you: "All quiet for both of you last night."
3. If your full low alarm goes unacknowledged through the whole local ladder while the radar confirms you are in the room, your buddy is alerted with one job: make a loud noise happen (a call through the relay), and if that fails, trigger your pre-written emergency script.
4. If you also opted into the hub, verified T1D volunteers anywhere in the world can see the pseudonymous listing, claim it exclusively, and do the same.
5. The morning after, the AI writes the close-out both ways: usually "all quiet"; after a real event, "Sam's okay. Your call at 3:12 woke him; he treated and recovered."

3.4 Pitch lines

- "You sleep. Someone's got you."
- "Somewhere on Earth, someone is awake for you. The sun never sets on the network."
- "The T1D community keeps watch over itself."
- "Only the watchable can be watchers."
- "A wasted call is thirty seconds of 'thanks for checking.' A missed emergency is the thing everyone here is afraid of. We know which mistake to prefer."

4. How It Ties to the Challenge, Criterion by Criterion

Meta criterion	How Night Buddy answers it
Meaningfully strengthens human connection	It creates relationships that did not exist (stranger pairs matched across the world), sustains them at near-zero daily cost (the AI carries the "all quiet" ritual), and gives them stakes no social app has: the person you were matched with may one day wake you out of a dangerous low. Even its false positives are connection: a needless check-in call from a peer is

	thirty seconds of warmth, not noise.
AI essential and well integrated	Three places, none decorative. (1) The 30-minute forecast is why a human backstop has time to matter at all. (2) The escalation judgment, when to wake a second human, fuses alarm state, elapsed time, presence, and re-arm history; a dumb timer would either cry wolf weekly (killing the buddy bond through alarm fatigue) or wait too long. (3) The narrative layer writes the daily "all quiet" and the morning-after story, carrying the relationship's ambient maintenance so the humans only talk when there is something human to say.
Originality	CGM sharing exists (raw data streams to people you already know). Volunteer-responder networks exist (PulsePoint, for cardiac arrest). Nobody has combined them: a peer-matched, AI-mediated overnight watch network for a chronic condition, with a claim-based community hub. The mirror-buddy insight (match across time zones so your watcher is awake) appears to be novel on its own.
Strength of the working demo	The demo beat is a physical device escalating in a dark room and a phone across the table lighting up, a claim, a brokered call, and the device room filling with sound. It is the most filmable thing the team owns, and the live version reuses core-build machinery (alarms, presence, relay, pairing).

One framing rule for the write-up and pitch: Night Buddy is a social product with a safety spine, not a medical device with a chat feature. Lead with the relationship (two strangers, one promise), let the ladder be the plumbing under it.

5. The Buddy System: Opting In, Matching, Daily Life

5.1 Joining the network

1. Eligibility: adults with T1D. The membership card is a connected CGM feed, weeks of real glucose data behind an account. This is automatic to check, essentially impossible to fake cheaply, and it enforces the community rule in one sentence: only the watchable can be watchers.
2. Opt-ins are separate and independent: (a) have a buddy, (b) be a watcher for your buddy, (c) appear on the hub when in distress, (d) volunteer on the hub. Each is its own toggle; any can be revoked instantly.
3. At sign-up the patient writes their emergency script: who to call if I cannot be woken, in what order, with what context. The script is theirs; volunteers execute it, never improvise medical decisions.
4. Buddy pairing uses the Rounds pairing machinery unchanged: QR, key exchange, matching 4-digit confirmation codes, PIN-gated confirm, revocable from either side. For matched strangers, the relay brokers the exchange and the codes confirm no key-swapping occurred.

5.2 Matching: twin or mirror

The person chooses what they want from the pairing:

- **Mirror buddy (safety-optimal).** Matched roughly 10 to 14 hours away. Your 2 AM is their 2 PM: your watcher is awake, phone in hand, mid-day. The weakest link in any backstop chain, "will they wake up," disappears. Reciprocity is preserved exactly: each covers the other's night during their own day, and neither ever loses sleep. Pitch: "somewhere on Earth, someone is awake for you."
- **Twin buddy (connection-optimal).** Matched in your own time zone. Shared rhythm, same mornings, and with two devices the wake chain runs room-scale in both directions (section 8). More likely to become an actual friendship; slightly weaker as a 3 AM backstop, since the watcher is also asleep.

A person can hold both kinds. Day-one matching is schedule compatibility plus stated preference; AI matching on management style and alarm patterns is roadmap, and honestly labeled as such. Either side can decline a proposed match or re-match at any time; the AI writes the short first-name-only introduction.

5.3 Daily life: designed to cost five seconds

The reason CGM-sharing arrangements collapse is burden. Night Buddy is engineered so the default day contains almost nothing: one AI-written morning line to each side ("All quiet for both of you," or "Rough night for Alex, two warnings, both handled"), tuned by the same no-invented-numbers validator as every other Irin narrative. No feeds, no dashboards, no chat obligation. A lightweight reply ("tell Alex nice save") is relayed as a one-line note. The relationship's ambient maintenance is the AI's job; the humans surface only when a human is needed.

5.4 Find a Buddy, the signup flow (added 2026-09-25)

The Irin Buddy tab of the hosted app (irin-out-of-sleep-at-hackgt.tech, four tabs, gated by the PIN with no login) carries the whole flow, one screen per step, and stores the result in the relay's directory (MongoDB Atlas) under a username. A username is a handle; the membership card is still a CGM feed behind the account. The steps:

5. Opt in: do you want a personal buddy? Then either pick someone you know by username (a search box over the directory) or be matched.
6. The globe: a d3-geo orthographic globe spun with a finger; the political timezone under the finger highlights as it turns and shows its current time; tap to select or unselect, as many zones as you like, your own zone pre-selected. Shapes from the timezone-boundary-builder GeoJSON, simplified; the clock from Intl.DateTimeFormat with the zone's IANA name. Fallback if the shapes fight back: 24 UTC-offset bands with the same interaction.
7. Languages you speak, so the buddy can speak one of them.
8. Availability on a week grid, with 22:00 to 08:00 locked as asleep and not uncheckable. This is what lets someone who is asleep during your free hours be paired with you: their sleep is your watch, yours is theirs. Availability can also be typed as free text; Muse Spark parses it into the grid rows, which you confirm before they count (the scorer uses the rows).
9. The emergency contact: the number (encrypted by the relay before storage, never shown to anyone; the relay places the call) and the instructions, rough notes that Muse Spark turns into a calm ordered script (JSON fields; shown only to the live claim-holder).

5.5 Matching, the pipeline (added 2026-09-25)

One aggregation pipeline in the relay against the directory: candidates opted in as watchers, language overlap, a timezone in the requester's set, and availability that covers the requester's sleep window converted to UTC; score = hours of the sleep window covered + 2 if the candidate is a mirror (UTC offset 10 to 14 hours away) + 1 per shared language; the top three are offered, either side can decline or re-match, and declined candidates never return. The introduction and the why-this-match line (from the score parts) are written by Muse Spark through narrative.py, first names only, never a glucose value; the score itself is deterministic and an LLM never picks the buddy (decision 29). AI matching on management style and alarm patterns stays roadmap, honestly labeled.

6. The Escalation Ladder

Everything before the final rung already exists in core Irin. Worked example with the default numbers (all starting values, all timed on clock.py so replay accelerates them):

1. T - 30 min: forecast predicts a low; amber WARNING (patient ramp, phone notification).
Acknowledged? Episode usually ends here as a near-miss. This rung is why the human rungs below rarely fire.
2. T: actual low crosses the threshold; FULL alarm: red LEDs, loud speaker, phone alert.
3. T + 5 min: unacknowledged; escalation: strobe, maximum volume.
4. T + 10 to 12 min: still unacknowledged AND presence radar confirms someone is in the room (in-room-but-unresponsive is the dangerous case; an empty room means the bathroom, not an emergency): the BUDDY RUNG fires. The buddy gets the urgent alert; their one job is to make a loud noise happen: a brokered call to the patient's phone, which also triggers the device to chime a distinct pattern. If the patient opted into the hub, the listing opens at the same moment. (Added 2026-09-25: the alert is a voice. The watcher page, kept in the foreground with a screen wake lock while "on watch", loops an ElevenLabs clip, "Your buddy Chris is in trouble, his alarm has been unacknowledged for twelve minutes," until Call is tapped; the relay sends the same text and clip to the buddy over WhatsApp as the second channel. Additive, like every rung: it never delays or quiets the local alarm.)
5. T + 20 min (independent clock): the patient's own emergency script executes regardless of anything above: the pre-set contact is alerted. Nothing on the buddy or hub path can postpone this. A volunteer claiming a listing buys the patient an extra chance, never a delay.
6. Resolution: the episode closes when the CGM shows recovery or the patient acknowledges. The AI writes the close-out to everyone involved.

Design notes: the buddy rung inherits the re-arm philosophy (if acknowledged but still low 15 minutes later, the ladder re-climbs); Brain-only patients have no presence check, so their rung 4 condition is stricter (section 8); and the arithmetic of rung 1 matters for the pitch: because the forecast fires 30 minutes early, the typical distance between "first warning" and "emergency contact called" is about 50 minutes, which is what makes a human phone-call rung meaningful at all.

7. The Hub: The Community Watch List

7.1 What it is

An opt-in, worldwide list of currently unacknowledged severe lows, visible to verified T1D volunteers, who can claim one and act. Precedent to cite: PulsePoint and GoodSAM alert CPR-trained citizens to nearby cardiac arrests and are credited with real saves. The hub is that proven pattern with "nearby" replaced by "awake," because the intervention is a phone call, and with the volunteer pool defined by lived experience instead of training: every volunteer has personally been the person fumbling for juice at 2 AM.

7.2 Who can volunteer

Other T1Ds only. Verification is the connected CGM feed plus account standing (weeks of real data, clean claim history). This one rule solves three problems at once: verification without bureaucracy (no ID checks, no training certificates, automatically checkable), abuse resistance (a fake account requires maintaining a fake CGM stream for weeks; one abusive claim burns an expensive identity), and meaning (the call that wakes you comes from someone who knows exactly what you are about to feel: "drink something, I've been there" is a different call than a stranger reading a script). It also makes the hub perfectly mutual: every volunteer is also protected. No helper class, no helped class; a community trading watches across time zones.

7.3 The listing

Pseudonymous and content-light: first name, alarm state, elapsed time, confidence label. No location, no glucose values, no photo, no feed. Two confidence tiers, always visible: "device alarm unacknowledged 14 min, presence confirmed" (Bedside) vs "no response to phone alerts, unconfirmed" (Brain-only). Volunteers triage with their eyes open, and the list is ordered by urgency and confidence.

7.4 The claim (the anti-stampede mechanic)

1. A volunteer taps "I've got this." The relay grants an exclusive lease, starting value 3 minutes; everyone else sees "claimed, in progress" and cannot pile on. A second simultaneous tap gets a clean conflict response.
2. While the claim is live, and only then, the claim-holder sees the patient's emergency script and gets the brokered-call button. No phone numbers are revealed in either direction.
3. The claim-holder's actions in order: brokered call (rings the phone AND chimes the device), then, if no response, execute the script (call the pre-set contact).
4. If the lease expires unresolved, the listing reopens automatically and the ladder keeps climbing regardless. A claim never pauses anything (invariant, section 9).
5. Every claim is logged: who, when, what actions, outcome. The audit trail is the answer to every abuse question.

7.5 The treating status (the community's acknowledgment)

If the person is actually awake and handling it, a needless check-in call is not a failure, it is the community working: thirty seconds of "I'm good, thanks for checking" from a peer across the world. The cost asymmetry is stark, a wasted call versus a missed emergency, so the hub deliberately leans toward listing and lets status do the cleanup:

1. One giant button on the phone alert and app: "I'm treating." It must be reachable in a single thumb press from the lock-screen notification, because hypoglycemia impairs exactly the faculties a fiddly UI demands. If marking treating takes two steps, people mid-low will not do it and the status system silently fails.
2. Tapping it clears the listing, shows any claim-holder "marked treating 2 min ago" so they can stand down (or still call to say hey, which is allowed and is kind of the point), and logs the event.
3. Treating expires, exactly like re-arm: starting value 20 minutes. If glucose has not recovered when it expires, the listing returns at TOP urgency: "marked treating 22 min ago, still under 54." A person who said they were handling it and still is not recovered is the strongest distress signal the hub can produce (undertreated, or lost consciousness after the tap).
4. Resolution is glucose, not words: a listing closes on CGM recovery or a clean treating window, never merely on a volunteer's report.

7.6 Scale honesty

Say this before a judge does: a buddy pair works at $n = 2$; the hub needs thousands before the list is ever non-empty. Order of magnitude: if 1 in 50 network members has a hub-worthy event per month, a 10,000-person network produces about 200 listings a month, roughly 7 a night worldwide, a number a few hundred volunteers absorb without fatigue. So the product ladder is: buddy pairs are the product that works tonight; the hub is what the network becomes at scale. In the demo, the hub is a labeled simulation; its job in the video is to show where this goes.

8. Device Scenarios: Both, One, or Neither

The device is never the ticket into the network, on either side of any feature. It is always the thing that upgrades signal quality. One sentence covers it: "Anyone with a CGM can watch and be watched; the bedside device makes your cry for help unmistakable."

Scenario	How the watch works	Honest limits (shown in the UI)
Both have devices (full pair)	The buddy rung goes to the buddy's ROOM, not their phone: a distinct light pattern and tone meaning "not your low, your buddy's." Irin's core competence, reliably waking a human, runs in both directions. Presence on both sides makes routing smart: if the buddy's radar says nobody is home, skip them immediately and go to the next rung instead of alerting an empty room. Marketing: "a pair of devices watches a pair of strangers."	None beyond the ladder itself. This is the flagship tier and the twin-buddy showcase.
One device (asymmetric, the common case)	The device owner gets the full ladder (room alarms, presence-confirmed escalation). The Brain-only buddy participates by phone: urgent notification when the device owner is unresponsive, morning messages both ways. In reverse, the Brain-only person is still watched:	The Brain-only side's "no response" is ambiguous (no radar): silent phone, phone in the kitchen, and true unresponsiveness look alike. Their screen says so; the

	unanswered phone warnings escalate to the device owner, under stricter criteria.	device owner's says "response confirmed by presence." Confidence labels (device_confirmed / unconfirmed) on the buddy card; a card never shows a blank or greyed row (Rounds evidence tiers).
Neither has a device (Brain-only pair)	A pure software social product: two people with CGMs and phones, matched, watched by the forecast, exchanging AI-written mornings. The wake chain is phone-to-phone, weakest, which is precisely why MIRROR matching matters most here: an awake watcher does not need waking, so a mirror Brain-only pair is nearly as strong as a device pair during the night.	Phone alerts can be missed; listings are low-confidence; the treating status and stricter listing criteria (confirmed level 2, sustained, zero phone interaction) do the triage. This tier is what makes Night Buddy a Meta-scale product: any of the millions of CGM users can join tonight, free.

The growth loop this creates: you join with your phone because it is free; you spend three months being the person whose 3 AM call once mattered; you buy the device because you have watched what it does from the outside. The buddy relationship itself is the sales channel.

9. Safety Invariants

Same status as the Irin and Rounds invariants: never weakened, tested by name, added to CLAUDE.md, review.md, and audit.md if Night Buddy code is built.

1. The buddy and hub rungs are ADDITIVE ONLY: they never delay, quiet, replace, or gate any local alarm or the emergency-script clock. The T + 20 emergency countdown runs independently of claims, calls, and statuses.
2. A claim never pauses the ladder. It buys the patient an extra chance, never a postponement.
3. Volunteers execute the patient's own pre-written script; the product never asks or allows them to give medical advice.
4. No glucose values, locations, or contact details ever appear on the hub. Contact is brokered by the relay only.
5. The emergency script is visible only to the current claim-holder, only while the claim is live.
6. Hub participation (watchable) and volunteering (watcher) are separate opt-ins, each revocable instantly; a revoked pairing deletes keys on both sides.
7. Only CGM-verified T1D accounts in good standing can see the hub list. Claim history is audited.
8. The treating button is one thumb press from the lock screen. (A UX rule promoted to invariant because its failure mode is silent.)
9. Listings are honest about confidence: Bedside-confirmed and Brain-only-unconfirmed are always visually distinct.

10. Demo/replay events are badged DEMO everywhere and never create real listings or real calls (extends Irin invariant 1).

10. Privacy Design

- The blunt statement of the risk, so the design is taken seriously: a naive public list of currently-unconscious vulnerable people would be a predator's directory. Every rule below exists to make the honest version.
- Pseudonymity: first names only, no location, no glucose data, no photos, anywhere in buddy or hub surfaces.
- Brokered contact: the relay connects calls; personal numbers never cross the network in either direction.
- Gated visibility: the hub list is visible only to verified volunteer accounts (CGM-backed, in standing), never public, never crawlable.
- Data minimalism: a buddy sees no feed, no dashboard, no history: only alerts that concern them and the daily narrative line.
- Everything E2E where person-to-person: buddy messages and alerts use the same crypto_box channel as Rounds cards; the relay stores ciphertext for content and sees only routing metadata. The WhatsApp message (decision 29) carries only the first name, the elapsed minutes and the watch URL, never a glucose value (relay invariant).
- Consent granularity: four separate toggles (buddy, watcher, hub-watchable, hub-volunteer), each reversible in one tap, with the other side notified ("sharing ended").
- Audit: every claim and every brokered call is logged for abuse review.

11. Technical Implementation

11.1 What is reused (the reason this is feasible at all)

Existing machinery	Role in Night Buddy
Alarm engine + escalation + re-arm (core)	Rungs 1 to 3 of the ladder, unchanged. The buddy rung is one new transition observed from outside.
Presence radar + AlarmEvent recorder (core + Rounds)	The "in room but unresponsive" condition; presence_during on every event; smart routing on the buddy side.
Relay + E2E crypto + pairing (Rounds)	The entire delivery layer: buddy pairing is the same handshake as doctor pairing; alerts and messages are crypto_box payloads; the hub is new endpoints on the same service.
Narrative pipeline + validator (core + Rounds)	The morning "all quiet," the introductions, and the event close-outs, with the same no-invented-numbers rule.
Replay engine + scenarios + DEMO badging (core)	Drives the demo: a scenario night triggers the real ladder at 60x on stage.
PIN gate, clock.py, state_snapshot rules (core)	All new mutating endpoints and timers follow the existing rules.

11.2 What is genuinely new

1. The buddy rung: one addition to the escalation path, implemented as an observer of alarm state (like the AlarmEvent recorder), never inside alarm.py's decision logic. Named test: full alarm, presence home, unacknowledged through two cycles → assert buddy alert fires; assert it never fires when Away; assert the T + 20 script clock is unaffected by claims.
2. Hub collections and endpoints on the relay: listings, claims with lease expiry, treating status, audit log. Roughly: POST /v0/hub/listing (device-originated), GET /v0/hub/list (volunteer-gated), POST /v0/hub/claim (lease; clean conflict on race), POST /v0/hub/treating, POST /v0/hub/resolve, GET /v0/hub/audit. The claim is ~20 lines of lease logic; documents on Atlas with a TTL index as the janitor.
3. Buddy alert delivery: two channels at the buddy rung, plus the honesty line. (1) The watcher page at watch.irin-out-of-sleep-at-hackgt.tech, kept in the foreground with a screen wake lock while on watch, receives the BuddyAlert over the WebSocket and loops audio_url, an ElevenLabs clip rendered once by Irin Cloud and cached. (2) The relay (relay/notify.py) sends the registered buddy a WhatsApp text through the WhatsApp Cloud API ("Irin: your buddy <first name> is in trouble. The alarm has been unacknowledged for N minutes. Open <watch URL>") and the same clip as an audio message (type audio, link on cloud.irin-out-of-sleep-at-hackgt.tech), via POST https://graph.facebook.com/<version>/{PHONE_NUMBER_ID}/messages; never a glucose value in the message (relay invariant); free-form messages only inside WhatsApp's 24-hour window after the buddy last messaged the number (each buddy sends "hi" during setup and before the demo), an approved template outside it; WHATSAPP_TOKEN and WHATSAPP_PHONE_NUMBER_ID on the server only. Honesty: a locked phone cannot be rung through silent from a web page at all. The demo is the wake-locked watcher page plus the WhatsApp message; the production answer is a native app with Critical Alerts, or a call (the WhatsApp/Twilio calling API, playing the same voice) from a number the buddy saved with Emergency Bypass on iPhone or "allow in Do Not Disturb" on Android, which rings through silent today with no app store.
4. The treating status: one lock-screen-action button, one endpoint, one expiry timer on clock.py, one "returns at top urgency" rule.
5. Matching: the aggregation pipeline of 5.5 in the relay against the Atlas directory is the build; pre-matched demo accounts are only the seeded fallback if the relay is unreachable. The twin/mirror preference is a field, and the AI-matching story is roadmap.
6. UI (Justin): buddy alert screen (huge, single-purpose), hub list with confidence labels and claim button, treating button on the alert notification and app home, buddy status card (pairing, revoke, morning message), and the split-screen demo arrangement. Added 2026-09-25: the buddy alert screen is the plain watcher page at watch.irin-out-of-sleep-at-hackgt.tech; everything else lives in the app's Irin Buddy tab (React), including the Find a Buddy screens of 5.4.
7. Contracts additions (one announced commit, if built): BuddyLink, BuddyAlert, HubListing (status: open | claimed | treating | resolved; confidence: device_confirmed | unconfirmed; claim_expires_at), HubClaim, EmergencyScript, plus WS types buddy_alert, hub_update, treating_set. Added 2026-09-25: audio_url on BuddyAlert; UserProfile and BuddyMatch for the directory and the aggregation's result.

12. Tech Stack

Piece	Technology	Status
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Ladder, alarms, presence	Existing core build (FastAPI, clock.py, HAL, LD2410B radar)	Existing
Delivery + crypto + pairing	The Rounds relay (FastAPI on the Vultr box behind Caddy, documents on MongoDB Atlas), PyNaCl / tweetnacl-js, QR pairing	Existing (Rounds; R1-R11 landed with a full day left)
Hub	New endpoints on the same relay; listings, claims, treating status and the audit log as documents on MongoDB Atlas, a TTL index as the janitor	NEW (small)
Buddy alerts	The watcher page at watch.irin-out-of-sleep-at-hackgt.tech (plain HTML, screen wake lock while on watch) receiving the BuddyAlert over a WebSocket and looping the ElevenLabs clip, plus the WhatsApp Cloud API message from relay/notify.py (text and the clip as audio; never a glucose value)	NEW (small)
Urgent brokered call	Telephony API (e.g. Twilio Voice): relay-initiated call, no numbers exposed	NEW (medium; demo-optional)
Matching	One aggregation over the Atlas directory (5.5): language overlap, timezone set, sleep-window coverage, mirror bonus, top three; pre-matched demo accounts only as the seeded fallback if the relay is unreachable	NEW (small)
Hosting	One Vultr VPS, Docker Compose (Caddy, relay, cloud), HTTPS on the .tech domain irin-out-of-sleep-at-hackgt.tech; the same compose file on Chris's laptop is the single-origin fallback	NEW (decision 27)
Relay storage + directory	MongoDB Atlas (M0): profiles, buddy links, hub listings with TTL indexes, matches, device pairings; the matching pipeline is one aggregation	NEW (replaces SQLite)
The voice	ElevenLabs text to speech, rendered once and cached: the alert loop, the script read aloud, Mom's voice note	NEW
The app	The hosted four-tab app at irin-out-of-sleep-at-hackgt.tech, a Vite + React + TypeScript + Tailwind build served by Caddy (Irin Buddy tab: buddy card, treating button, hub list with confidence labels and claim, Find a Buddy with the d3-geo globe in a React component); the watcher page stays plain HTML	NEW (replaces the Pi-served vanilla PWA)
Narratives	narrative.py's chain: Backboard (memory per buddy pair, model routing) → the Meta Model API direct (muse-spark-1.3 for the introduction, the why-this-match line, the emergency script and the availability parse) → the direct Claude call → deterministic template, the same no-invented-numbers validator after every link;	NEW

	NARRATIVE_ROUTING picks the provider per task	
Meta	Meta Model API (https://api.meta.ai/v1 , OpenAI-SDK compatible, META_MODEL_API_KEY) for the Buddy narratives; WhatsApp Cloud API (WHATSAPP_TOKEN, WHATSAPP_PHONE_NUMBER_ID) as the second alert channel; optional Facebook Login (VITE_FB_APP_ID, /me/friends) for a buddy you already know, only if Justin has time, cut first	NEW (decision 29)

Honest note on phone alerts: a web page cannot be trusted alone for a 3 AM life-safety alert (a locked, silenced phone; focus modes; OS throttling). The design compensates four ways: the mirror-buddy mode makes the watcher awake by construction; the WhatsApp message is a second channel to the buddy; the urgent rung is a real phone call; and the patient's own emergency script fires on its independent clock no matter what. Say this proactively; it turns a weakness into evidence of design maturity.

13. Extra Work Required and Build Tiers

13.1 Prerequisites

Night Buddy code starts only after (1) core Irin runs end to end (the freeze rule) and (2) R1-R11 landed with a full day left, because the relay, pairing, crypto, and narrative machinery ARE the Night Buddy delivery layer. Skipping ahead would mean building that layer twice.

13.2 Three build tiers

1. VIDEO-ONLY (always achievable): no new code. The video stages Night Buddy with the mock HAL on two laptops and a phone, clearly labeled concept. Cost: half a day of Slavik's video time plus script. The Meta entry is then Family Story (built) + Night Buddy (concept climax).
2. DEMO TIER (the target if Rounds lands early): the Find a Buddy flow and the matching aggregation (5.4, 5.5; pre-matched demo accounts only as the seeded fallback if the relay is unreachable), buddy rung observer + named tests, the wake-locked watcher page on a teammate's phone looping the voice plus the WhatsApp message, mock hub page with live claim + treating + reopen-on-expiry, morning narrative. Estimate: about one day of Chris, half a day of Justin, on top of everything else. No telephony.
3. FULL TIER (only if the weekend is going suspiciously well): the production alert channel (a native app with Critical Alerts, or a call through the WhatsApp/Twilio calling API), the brokered Twilio call to the patient, hub audit view, optional Facebook Login for a buddy you already know. Estimate: one to two more days combined. Realistically post-hackathon.

13.3 By owner (demo tier)

Owner	Work
Chris	Buddy rung observer + tests; hub endpoints (listing, claim lease, treating, resolve) on the relay; the WhatsApp send (relay/notify.py); treating expiry on clock.py; contracts additions; narrative prompts
Justin	Buddy alert screen; hub list page with confidence labels + claim; treating button on the alert; buddy status card and the Find a Buddy screens in the app's Irin Buddy tab

George	Nothing required. Optional: a scenario tuned so the ladder timing demos cleanly at 60x
Slavik	The video (owns it regardless of tier); staging the split-screen; the second "device" is the mock HAL on a laptop

13.4 The scope warning, stated plainly

The team is already carrying the core device, the ML pipeline, and the two-card Rounds build. Night Buddy's demo tier is small ONLY because of what it reuses, and it is still the first thing to cut. The decision rule, agreed now: unless R1-R11 landed with a full day left, Night Buddy is video-only, no debate at 3 AM. The Meta entry is strong either way, because Family Story rides entirely on the morning-report pipeline and stands alone.

14. The Demo and the Video

14.1 The 2-to-3-minute video arc

1. 0:00 The problem, first person: Chris, alone, phone on the nightstand, the one fear every T1D adult living alone shares. One line on Dexcom Follow: "the only fix today is giving someone your medical data forever."
2. 0:30 The promise: matched with a stranger. The AI introduction. The morning ritual: "All quiet for both of you." Show a week of five-second mornings.
3. 1:00 The night it matters (split screen): left, the device escalating in a dark room, nobody stirring, radar dot showing someone present; right, a phone across the world lighting up in daylight (mirror buddy). The claim. The brokered call. The dark room fills with ringtone and chime. Movement. Juice.
4. 1:45 The morning after, both screens: "Sam's okay. Your call at 3:12 woke him; he treated and recovered." Beat of silence.
5. 2:00 The hub, labeled concept: the watch list, a claim, the treating button, "only the watchable can be watchers." One line of scale: the sun never sets on the network.
6. 2:20 Close on the thesis: "AI does not replace the person who checks on you. It finds them, and it knows when to wake them."

14.2 Live demo (demo tier)

Device on the table as the patient; a teammate's phone as the mirror buddy with the watcher page open (wake lock); replay scenario drives a real unacknowledged low at 60x; the buddy alert lands on the page and as a WhatsApp message; the teammate claims and taps the brokered-call mock; the device chimes; acknowledge; the morning message renders. Total: about 45 seconds, all real code paths, DEMO badges everywhere.

15. The Write-Up Skeleton (Meta Requires One)

1. WHO IT IS FOR: the T1D adults who live alone (and their 3 AM fear with a community name), plus the wider CGM-wearing community as watchers. Family Story (the companion feature) covers the

families; Night Buddy covers the people families cannot reach: strangers to each other, until matched.

2. **HOW IT STRENGTHENS CONNECTION:** it mints new relationships with real stakes; the AI carries the daily maintenance so the bond costs five seconds a night; its false positives are check-in calls from peers, warmth by design; and the hub turns a scattered global community into a mutual watch. Cite the T1D peer-support literature for outcomes.
3. **WHY AI IS ESSENTIAL:** the forecast creates the time in which a human backstop is possible; the escalation judgment decides when waking a second human is justified (the anti-alarm-fatigue rule that keeps the bond alive); the narrative layer sustains the relationship between events. Remove any of the three and the product collapses into either a noisy pager or a raw data feed, both of which already exist and both of which fail.
4. **SAFETY POSTURE** (pre-empt it): additive rungs only; the patient's own script; volunteers never advise; resolution by glucose; the liability question answered honestly ("production needs duty-of-care review; v1 frames the buddy as a friend who happens to be far away, not a service guarantee").

16. Q&A and Risks

Question / risk	Answer / mitigation
Isn't this Dexcom Follow?	Follow shares a raw data stream with people you already know, forever. Night Buddy shares nothing by default, matches you with someone new, and transmits exactly one thing: the moment you need a human. Opposite designs.
What if the buddy sleeps through it?	Mirror matching makes the watcher awake by construction; twin pairs with devices get room-scale waking; and the patient's emergency script fires on an independent clock regardless. The buddy is a bonus rung, never the last line.
What if a volunteer does nothing after claiming?	Leases expire and the listing reopens; the ladder never paused; claim history is audited and bad accounts lose hub access.
What about pranks or predators?	The list is visible only to CGM-verified T1D accounts in standing (weeks of real data to fake), entries are pseudonymous and content-free, contact is brokered with no numbers exposed, and everything is audited.
Is a stranger responsible for my safety?	No. They execute your own pre-written script, and every rung below and above them fires on schedule regardless. Duty-of-care review is named as production work.
False alarms will burn volunteers out.	The rung fires only after the full local ladder fails with presence confirmed, rare by construction; treating status clears benign cases in one tap; and this community reads a needless check-in as care, not noise. The math: even a 10,000-person network produces only a handful of listings a night worldwide.
The pool is too small for the hub.	Conceded and stated first: pairs work at $n = 2$; the hub is the network's mature form. The demo labels it as such.

Why T1D-only volunteers?	Verification (the CGM feed is the membership card), abuse economics, and meaning: the call comes from someone who has been there. T2D insulin users are a settings flag away when the community decides.
Phone push is unreliable.	Agreed, and said on stage: a web page cannot ring through a locked, silenced phone. The demo keeps the watcher page open with a wake lock and loops the voice, and the relay sends the same alert as a WhatsApp message (text plus the clip). Production: a native app with Critical Alerts (what Dexcom has), or a call from a number saved with Emergency Bypass / allowed in Do Not Disturb (the WhatsApp/Twilio calling API), playing the same voice; and the mirror watcher is awake anyway; and the emergency script fires on its own clock regardless.
Scope: you are building three products.	Night Buddy demo tier exists only because it reuses the Rounds delivery layer, and the pre-agreed rule cuts it to video-only if Rounds is late. Family Story alone carries the Meta entry if needed.

17. Open Items

1. Telephony (Twilio) account + a test number, only if the full tier becomes plausible; otherwise skip.
2. Check the wake-locked watcher page and the WhatsApp message on the team's actual phones (each buddy phone has messaged the test number inside the 24-hour window) before the video narration promises anything; production = native Critical Alerts or a call (WhatsApp/Twilio calling API).
3. Pick the demo phones: the buddy phone should be a teammate's, pre-provisioned, on the hotspot fallback with the watcher page open and a "hi" already sent to the WhatsApp test number.
4. Added 2026-09-25: the domain is irin-out-of-sleep-at-hackgt.tech (registered with the MLH .Tech code, Active on Cloudflare); the seven accounts the flow needs are in docs/OPEN_ITEMS.md decisions 27 and 29 (the .tech domain, Vultr, Atlas, Tiger Cloud, ElevenLabs, Backboard, the Meta developer app).

17A. Hosting, the App, and the Sponsor Integrations (added 2026-09-25)

Everything Night Buddy needs in the cloud runs on one Vultr VPS from `deploy/docker-compose.yml` behind Caddy with automatic HTTPS on the team's domain, `irin-out-of-sleep-at-hackgt.tech`: the relay (courier, hub, buddy directory and matching; documents on MongoDB Atlas; never a glucose value) and Irin Cloud (the patient's own service: time series on Tiger Cloud, the My Irin dashboards, the family rollup, and ElevenLabs audio rendering). The watcher page lives at `watch.irin-out-of-sleep-at-hackgt.tech`, the app at `irin-out-of-sleep-at-hackgt.tech`, the relay at `api.irin-out-of-sleep-at-hackgt.tech`, Irin Cloud at `cloud.irin-out-of-sleep-at-hackgt.tech`. If Vultr or the venue network fails, the same compose file runs on Chris's laptop with local mongo and timescaledb and serves everything on one origin, `http://<laptop-ip>` (the app's dist rebuilt with `VITE_*` values pointing at it, the Pi's `APP_ORIGIN` set to it; rehearsed once at home; `chris.md`, R6 fallback, is the one statement), and phones on the hotspot open that address.

The app has four tabs: Irin Device (talks to the paired Pi through its tunnel at device.irin-out-of-sleep-at-hackgt.tech after the kiosk's QR + 6-digit code is redeemed through the relay, POST /v0/device/pair; a DevicePairing, separate from doctor and buddy pairings), My Irin (dashboards; the Buddy dashboard shows nights covered, the all-quiet streak, alerts sent and received, and never a glucose value), Irin Buddy (5.4 and 5.5 above, the buddy card, the one-press treating button), Irin Rounds. No login; the PIN gates it.

Honesty about the locked phone, in the pitch and the write-up: a web app cannot ring through silent or Do Not Disturb; only a native app with Critical Alerts, or a call from a number the buddy has saved with Emergency Bypass (the WhatsApp/Twilio calling API), can. The demo shows the foreground page with the voice loop plus the WhatsApp message (text and the clip as an audio message) and says exactly that.

Decision 29 (2026-09-25), the Meta pieces. The Meta Model API (<https://api.meta.ai/v1>, OpenAI-SDK compatible through the openai package with base_url; key META_MODEL_API_KEY on the Pi and on the server; openai added to backend/requirements.txt and relay/requirements.txt) runs muse-spark-1.3 for what Muse writes in Irin Buddy: the introduction line and the why-this-match line (from the score parts; first names only, never a glucose value), the emergency script (rough notes → a calm ordered script, JSON fields), free-text availability parsed into the week-grid rows the user then confirms, and the Family Story. narrative.py's chain is four links, Backboard → Meta direct → direct Anthropic → deterministic template, with the no-invented-numbers validator after every link; NARRATIVE_BACKEND keeps backboard | anthropic | template and NARRATIVE_ROUTING picks the provider per task, eight tasks: clinician_card and morning_report → anthropic (a Sonnet-class Claude); buddy_line and family_story → openrouter with the Muse Spark name if Backboard's OpenRouter carries it, else meta / muse-spark-1.3; match_explanation, emergency_script and availability_parse → meta / muse-spark-1.3 direct with structured output; second_opinion → openai. The matching score stays deterministic (the Atlas aggregation of 5.5); an LLM never picks the buddy. The WhatsApp Cloud API is the second alert channel (11.2 item 3; WHATSAPP_TOKEN and WHATSAPP_PHONE_NUMBER_ID on the server only), and optional Facebook Login (VITE_FB_APP_ID in frontend/app/.env.production; /me/friends returns friends who also use Irin) covers a buddy I already know, only if Justin has time, cut first. WhatsApp payloads and Muse prompts never carry a glucose value; /review and /audit grep relay/notify.py and the buddy prompts for mgdl.

Python 3.14 everywhere (the Pi through uv). Cursor runs beside Claude Code under the same constitution; nothing in this document changes for it.

18. Relationship to the Other Two Entries

One architecture, three stories, told consistently at every table:

- Main judging: Irin, the bedside guardian. The device and The Save. (Irin spec.)
- Impiricus: Irin Rounds, the guardian reports to the doctor. Two cards over the blind courier. (Rounds spec.)
- Meta: Family Story + Night Buddy, the guardian connects the people around the patient: family gets the story instead of the data; peers get each other. (This document + the Family Story feature riding the morning-report pipeline.)

Everything hangs off the same spine: the forecast, the ladder, presence, the relay, the narrative-with-validator. Each sponsor sees a different door into the same house, which is also the honest answer if any judge asks whether the project is three hacks stapled together: it is one system, and these are its three audiences: the sleeper, the doctor, and the people who care about the sleeper.